

TidyTuesday Week 37

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Wednesday 9th September, 2026

Contents

1	Weekly Summary	2
2	Date: 2026-09-07	4
2.1	Raw Regression	4
2.2	Detrended Regression	6
3	Date: 2026-09-08	8
3.1	Raw Regression	8
3.2	Detrended Regression	10
4	Date: 2026-09-09	12
4.1	Raw Regression	12
4.2	Detrended Regression	14

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we shove two economic time series that have never met into a blender and act surprised when soup comes out. This week our random number generator, in its infinite wisdom, decided to introduce Lithuanian pandemic chatter to U.S. federal borrowing, coal emissions to congenital anomaly price indexes, and the market value of the entire federal debt to a niche corner of business equipment leasing. Truly, the pairings of legends. Let's see what falls out.

Monday gave us the "Discussion About Pandemics Index for Lithuania" versus U.S. "Borrowing from the Public," and honestly I need everyone to sit down for this one because the relationship actually survived detrending. The raw regression found a significant link, sure, but so does everything that trends upward while looking vaguely important. The shocking part is that after we stripped out each series' own time trend, the connection barely flinched – p-value still comfortably significant, R-squared drifting from about 0.20 down to a still-respectable 0.17. In other words, Lithuanians chatting about pandemics wobbles in step with how much the U.S. Treasury borrows, even after the boring shared drift is removed. Do I have a mechanism for this? Absolutely not. Am I going to pretend the Baltic pandemic discourse is a leading indicator of American fiscal policy? Only every day for the rest of my life.

Tuesday's coal emissions versus congenital anomaly medical price index was the real over-achiever, at least at first glance: a raw R-squared of 0.61, which for this project is basically a marriage proposal. And in a rare plot twist, the detrended version held up too, with the R-squared settling around 0.56 and the p-value staying tiny. So we have two things – coal being burned and the price of treating congenital anomalies – moving together in a way that outlasts their trends. Before anyone drafts a public health manifesto, please recall that both series are annual, short, and were selected by a script that also thinks Lithuania is a Treasury analyst. The condition-number warnings are the software's polite way of clearing its throat and reminding us not to take this too seriously. It's a genuinely strong-looking result, which mostly proves that strong-looking results are exactly what randomness produces if you draw enough pairs.

Wednesday, mercifully, restored order to the universe. The Market Value of Gross Federal Debt versus business equipment leases posted a gorgeously significant raw regression – p-values with more zeros than a national deficit – and then coughed up an R-squared of a whopping 0.10, which is the statistical equivalent of a firm handshake from someone who then forgets your name. The detrended version stayed "significant" but explained even less, so this is your classic trend-mirage: two things that both crawled upward over decades, waving at each other across the

drift, with no meaningful relationship to show for it. As always, the standing disclaimer: these are randomly paired series, nobody is claiming causation, and spurious correlation isn't the bug here, it's the entire personality of the project. See you next week, when the robot pairs something even weirder and I once again pretend to be surprised.

2 Date: 2026-09-07

Series ID: WPDILTU

This series is titled Discussion About Pandemics Index for Lithuania and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: MTSMFBP133FMS

This series is titled Means of Financing: Borrowing from the Public and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2010-01-01 and the observation end date is 2026-07-01. The popularity of this series is 14.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MTSMFBP133FMS	R-squared:	0.199			
Model:	OLS	Adj. R-squared:	0.186			
Method:	Least Squares	F-statistic:	15.87			
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000176			
Time:	16:43:46	Log-Likelihood:	-893.46			
No. Observations:	66	AIC:	1791.			
Df Residuals:	64	BIC:	1795.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.886e+04	2.46e+04	1.985	0.051	-321.644	9.8e+04
value_fred_WPDILTU	1205.6312	302.613	3.984	0.000	601.092	1810.170
Omnibus:	51.230	Durbin-Watson:	2.093			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	275.412			
Skew:	2.139	Prob(JB):	1.57e-60			
Kurtosis:	12.047	Cond. No.	87.5			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

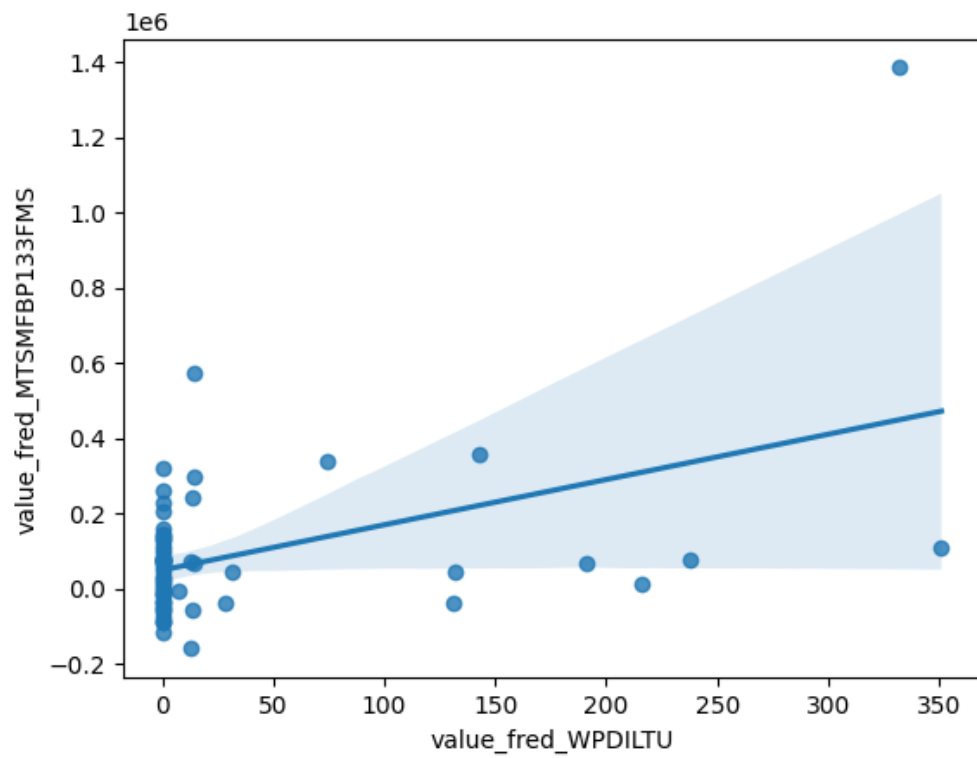


Figure 1: Raw Regression Plot for 2026-09-07

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MTSMFBP133FMS_detrended	R-squared:	0.174
Model:	OLS	Adj. R-squared:	0.161
Method:	Least Squares	F-statistic:	13.48
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000493
Time:	16:43:46	Log-Likelihood:	-893.06
No. Observations:	66	AIC:	1790.
Df Residuals:	64	BIC:	1794.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
const	5.051e-09	2.28e+04	2.22e-13	1.000	-4.55e+04	4.55e+04
value_fred_WPDILTU_detrended	1138.7087	310.105	3.672	0.000	519.204	1758.214

Omnibus:	52.012	Durbin-Watson:	2.129
Prob(Omnibus):	0.000	Jarque-Bera (JB):	302.597
Skew:	2.138	Prob(JB):	1.96e-66
Kurtosis:	12.579	Cond. No.	73.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

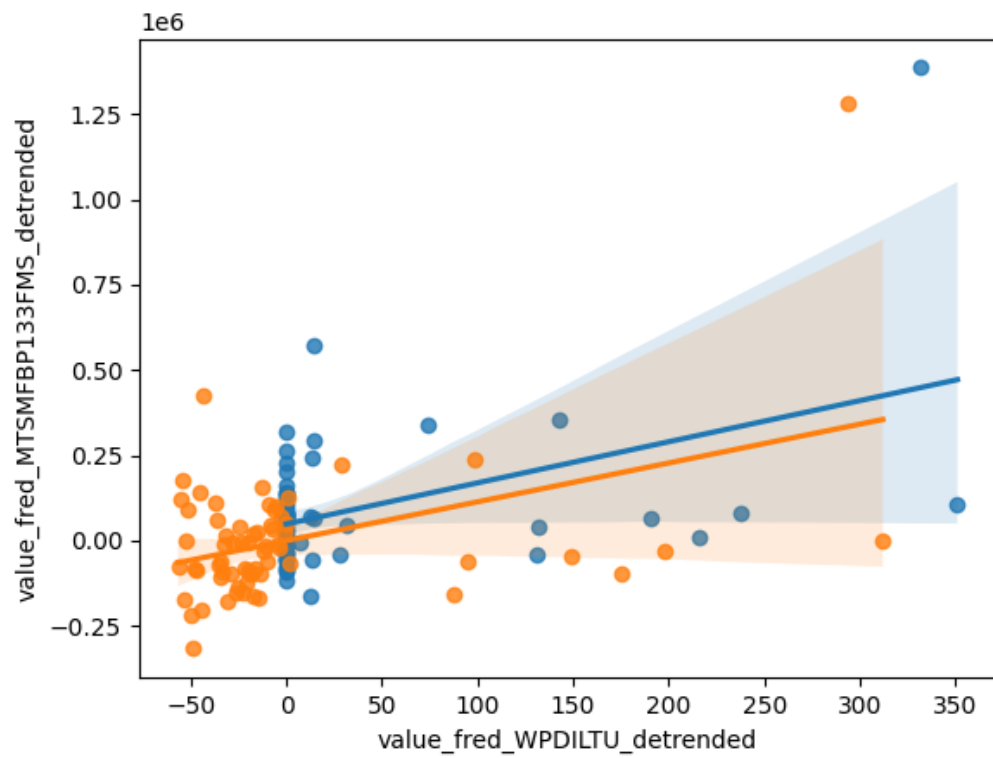


Figure 2: Detrended Regression Plot for 2026-09-07

3 Date: 2026-09-08

Series ID: EMISSCO2TOTVTTCOUSA

This series is titled Total Carbon Dioxide Emissions From All Sectors, Coal for United States and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1970-01-01 and the observation end date is 2021-01-01. The popularity of this series is 5.

Series ID: COGANOPIMEPS

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2009=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2020-01-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_COGANOPIMEPS	R-squared:	0.606
Model:	OLS	Adj. R-squared:	0.585
Method:	Least Squares	F-statistic:	29.22
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	3.24e-05
Time:	15:27:44	Log-Likelihood:	-111.42
No. Observations:	21	AIC:	226.8
Df Residuals:	19	BIC:	228.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-146.4574	51.777	-2.829	0.011	-254.827	-38.088
value_fred_EMISSCO2TOTVTTCOUSA	0.1533	0.028	5.406	0.000	0.094	0.213

Omnibus:	3.022	Durbin-Watson:	1.167
Prob(Omnibus):	0.221	Jarque-Bera (JB):	1.530
Skew:	0.625	Prob(JB):	0.465
Kurtosis:	3.429	Cond. No.	8.45e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.

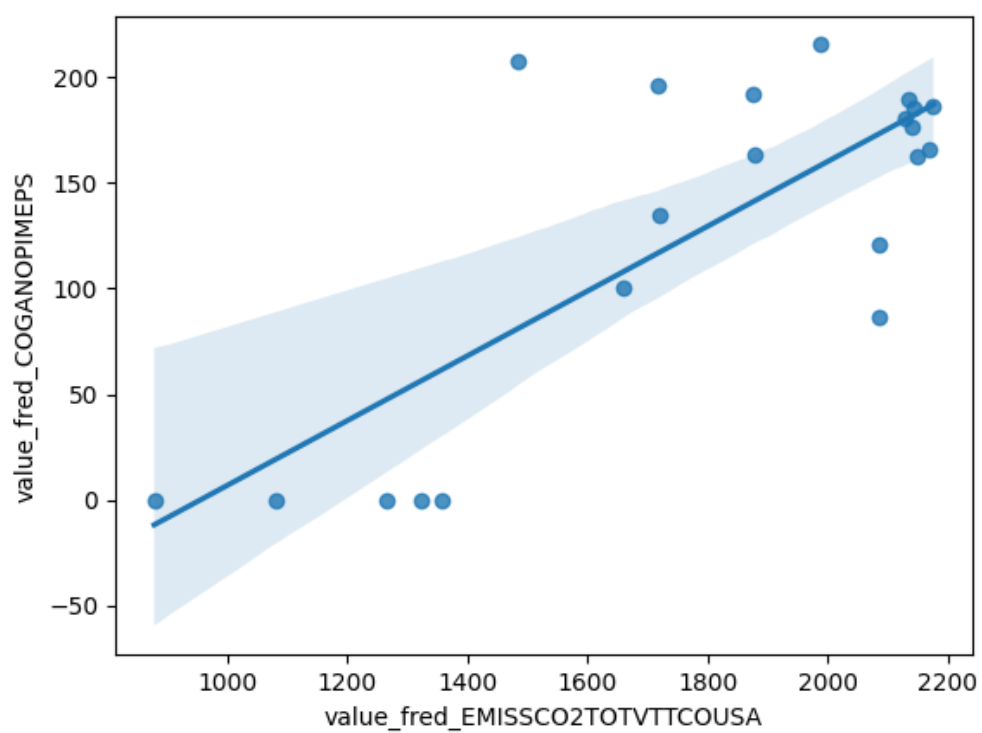


Figure 3: Raw Regression Plot for 2026-09-08

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_COGANOPIMEPS_detrended	R-squared:	0.559				
Model:	OLS	Adj. R-squared:	0.536				
Method:	Least Squares	F-statistic:	24.09				
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	9.76e-05				
Time:	15:27:44	Log-Likelihood:	-108.14				
No. Observations:	21	AIC:	220.3				
Df Residuals:	19	BIC:	222.4				
Df Model:	1						
Covariance Type:	nonrobust						
		coef	std err	t	P> t	[0.025	0.975
const		-6.734e-12	9.567	-7.04e-13	1.000	-20.023	20.023
value_fred_EMISSCO2TOTVTTCOUSA_detrended		0.3032	0.062	4.908	0.000	0.174	0.432
Omnibus:		4.692	Durbin-Watson:	1.541			
Prob(Omnibus):		0.096	Jarque-Bera (JB):	2.578			
Skew:		0.726	Prob(JB):	0.276			
Kurtosis:		3.915	Cond. No.	155.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

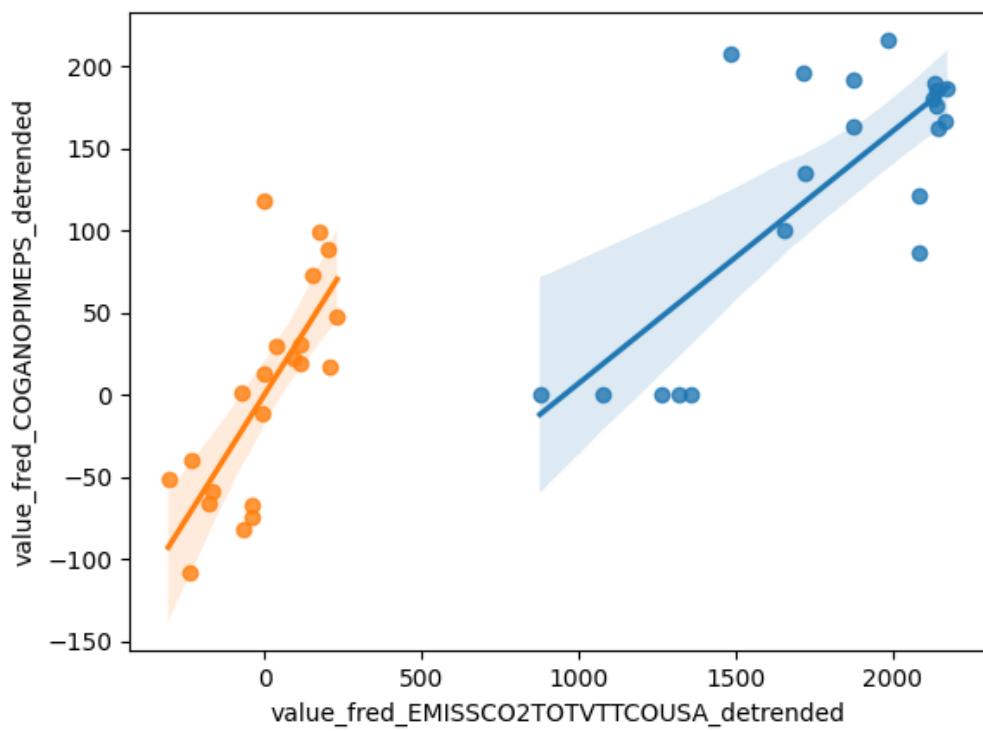


Figure 4: Detrended Regression Plot for 2026-09-08

4 Date: 2026-09-09

Series ID: MVGFD027MNFRBDAL

This series is titled Market Value of Gross Federal Debt and has a frequency of Monthly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1942-01-01 and the observation end date is 2026-07-01. The popularity of this series is 37.

Series ID: DTBTERNM

This series is titled Business Equipment Leases Owned and Securitized by Finance Companies, Level and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-06-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DTBTERNM	R-squared:	0.095
Model:	OLS	Adj. R-squared:	0.093
Method:	Least Squares	F-statistic:	45.33
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	5.34e-11
Time:	15:24:33	Log-Likelihood:	-5318.4
No. Observations:	433	AIC:	1.064e+04
Df Residuals:	431	BIC:	1.065e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.633e+05	4472.957	36.517	0.000	1.55e+05	1.72e+05
value_fred_MVGFD027MNFRBDAL	-1.7290	0.257	-6.732	0.000	-2.234	-1.224

Omnibus:	2749.157	Durbin-Watson:	0.021
Prob(Omnibus):	0.000	Jarque-Bera (JB):	48.079
Skew:	0.054	Prob(JB):	3.63e-11
Kurtosis:	1.371	Cond. No.	3.10e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.1e+04. This might indicate that there are strong multicollinearity or other numerical problems.

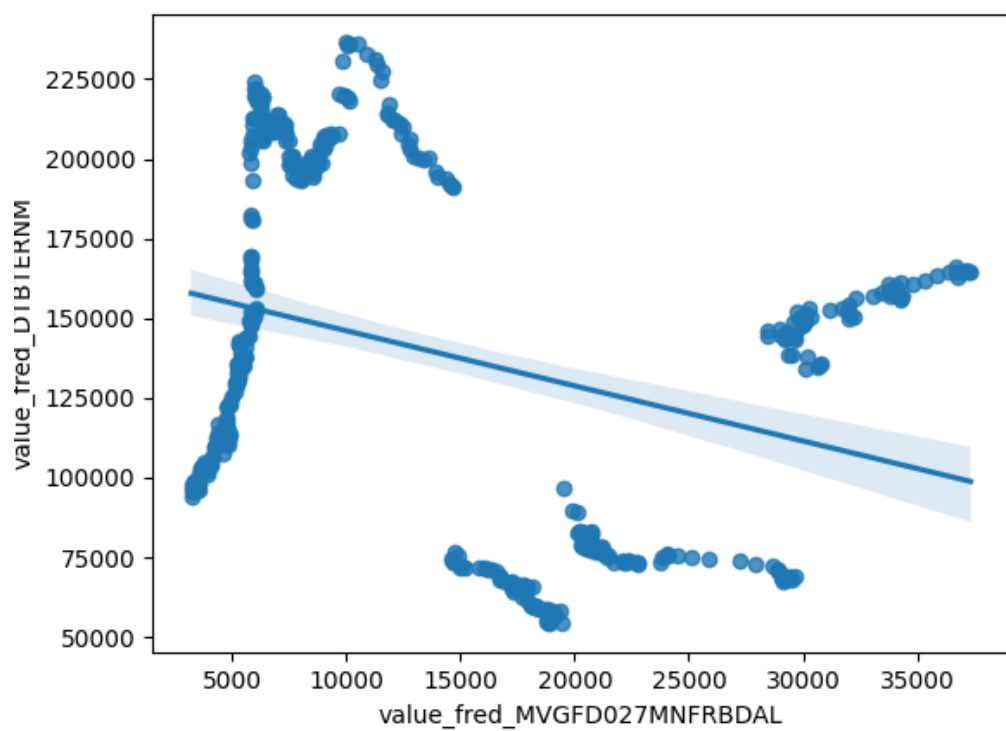


Figure 5: Raw Regression Plot for 2026-09-09

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DTBTERNM_detrended	R-squared:	0.116
Model:	OLS	Adj. R-squared:	0.114
Method:	Least Squares	F-statistic:	56.72
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	2.97e-13
Time:	15:24:33	Log-Likelihood:	-5302.8
No. Observations:	433	AIC:	1.061e+04
Df Residuals:	431	BIC:	1.062e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975
const	9.044e-09	2427.545	3.73e-12	1.000	-4771.300	4771.300
value_fred_MVGFD027MNFRBDAL_detrended	-6.1487	0.816	-7.531	0.000	-7.753	-4.543

Omnibus:	698.313	Durbin-Watson:	0.023
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33.214
Skew:	-0.174	Prob(JB):	6.13e-08
Kurtosis:	1.689	Cond. No.	2.97e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.97e+03. This might indicate that there are strong multicollinearity or other numerical problems.

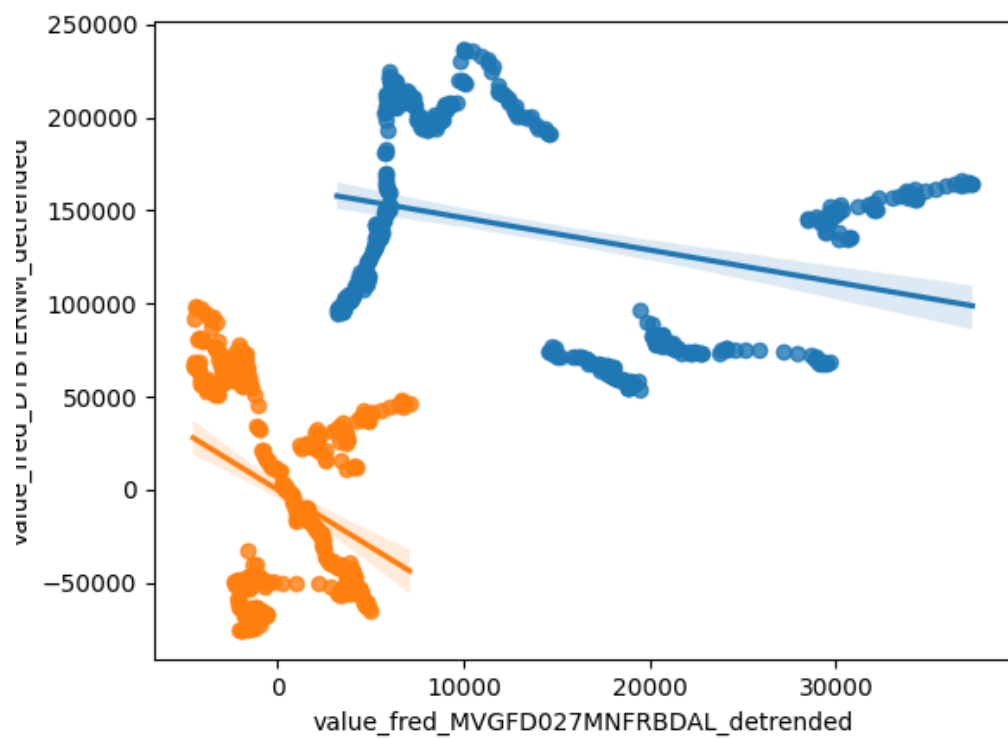


Figure 6: Detrended Regression Plot for 2026-09-09